

Chemical + ionic equations

Eg. Iron(III) sulphate and potassium hydroxide react to form potassium sulphate solution and iron II hydroxide.

$FeSO_{4(aq)} + 2KOH_{(aq)} \rightarrow Fe(OH)_{2(s)} + K_2SO_{4(aq)}$

Ionic $Fe^{2+}_{(aq)} + 2OH^-_{(aq)} \rightarrow Fe(OH)_{2(s)}$

1. $CuSO_4 + NaOH_{(aq)} \rightarrow Cu(OH)_{2(s)} + Na_2SO_4$

   $Cu^{2+}_{(aq)} + 2OH^-_{aq} \rightarrow Cu(OH)_{2(s)}$

2. $BaCl_{2(aq)} + NaSO_{4(aq)} \rightarrow NaCl_{(aq)} + BaSO_{4(s)}$

   $Ba^{2+} + SO_4^{2-} \rightarrow BaSO_{4(s)}$

Oxidation Number

rules:	oxidation number
Element	0
Simple ions (one element)	charge number
Hydrogen	+1
Hydrogen in metal hydride	NaH (+1 / −1)
Flarine	−1
oxygen	−2 unless bonded to f or in a peroxide

peroxide = $H_{+1}-O^{-1}-O^{-1}-H_{+1}$

eg. SO_2 $\quad S + 2(-2) = 0$

$S - 4 = 0$

$S = 4$

SO_4^{2-} $\quad S + 4(-2) = -2$

$S = 6$

Oxidation number recap

1

$\underline{Ba}Cl_2$	+2	
$\underline{Li}_2O$	+1	
$\underline{P}_2O_3$	+3	$P_2 - 6 = 0$
$\underline{C}Cl_4$	+4	$C - 4 = 0$
$\underline{C}O$	+2	
CrO_3	+6	
CrO_4^{2-}	+6	$Cr - 8 = -2$
$Cr_2O_2^{2-}$	+4	$Cr_2 - 8 = -2$

$\underline{S}O_3^{2-}$	+4	$S - 6 = -2$
$\underline{Br}O_3^-$	+5	$Br - 6 = -1$
$\underline{I}_2$	0	
$\underline{I}^-$	-1	
$\underline{I}O_3^-$	+5	
$\underline{N}O_2$	+4	
$\underline{N}_2O_4$	+4	
$\underline{P}O_4^{3-}$	+5	$P - 8 = -3$

$\underline{Cl}O^-$	+1	$Cl - 2 = -1$
$\underline{Cl}O_3^-$	+5	$Cl - 6 = -1$
$\underline{Cl}_2O_7$	+7	$Cl_2 - 14 = 0$
$F_2\underline{O}$	+2	$O - 2 = 0$
$H_2\underline{O}_2$	-1	
$\underline{S}_2O_3^{2-}$	+2	$S_2 - 6 = -2$
$\underline{S}_4O_6^{2-}$	+2.5	$S_4 - 12 = -2$
$\underline{Li}\,\underline{H}$	-1	

Ch 23.1

1. $Mg + Zn^{2+} \rightarrow Mg^{2+} + Zn$
   (0 → +2; +2 → 0)

   Mg = reducing agent

   Zn = oxidising agent

b) $2H_2SO_4 + Cu \rightarrow CuSO_4 + SO_2 + 2H_2O$
   (+2 S −8 → 2+S −8; S = 6 → S = 4; 0 → +2)

   S = reduced = oxidising agent

   Cu = reducing agent

c) $Fe^{2+} + 2H^+ + NO_2^- \rightarrow Fe^{3+} + H_2O + NO$
   (+2 → +3; +1; N − 4 = −1, N = 3; +2 −2)

   Fe = reducing agent

   N = oxidising agent

2. a) $6Fe^{2+} + Cr_2O_7^{2-} + 14H^+ \rightarrow 6Fe^{3+} + 2Cr^{3+} + 7H_2O$

   b) $3Zn + 2VO_2^+ + 8H_2^+ \rightarrow 3Zn^{2+} + 2V^{2+} + 4H_2O$

   c) $H_2O_2 \rightarrow O_2 + 2H^+ + 2e^-$

   $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$

   $5H_2O_2 + 2MnO_4^- + 10H^+ \rightarrow 5O_2 + 2Mn^{2+} + 8H_2O$

3. a) $2MnO_4^- + 6H^+ \rightarrow 5O_2 + 2Mn^{2+} + 8H_2O$

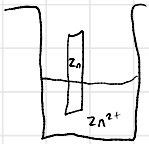
   b) $BrO_3^- + 5Br^- + 6H^+ \rightarrow 3Br_2 + 3H_2O$
   (Br − 6 = −1, Br = 5; −1; +1; 0; +1 −2; +1)

4. a) $MnO_2 + 4H^+ + 2e^- \rightarrow Mn^{2+} + 2H_2O$
   (+5 → +2)

   b) $NO_3^- + 2H^+ + 3e^- \rightarrow NO + 2H_2O$
   (H = 1 × 4; N = 1; O = 3; H = 2 × 4; N = 1; O = 2 × 3)

Ch 23.4

Voltaic cell = Converts chemical energy into electrical.
Half cell = Chemical species in a half equation.



No net transfer of e^-

$Zn^{2+}(aq) \mid Zn(s)$

$Fe^{3+}(aq) + e^- \rightleftharpoons Fe^{2+}(aq)$

More reactive metal release e^- and oxidised.

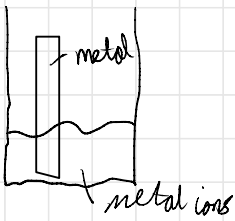
- electrode more reactive metal loses e^-, oxidised - negative electrode.
- less reactive metal gains e^-, reduced - positive electrode.

Standard electrode

$2HCl + Mg \rightarrow MgCl_2 + H_2$

$2CH_3COOH + Mg \rightarrow Mg(CH_3COO)_2 + H_2$

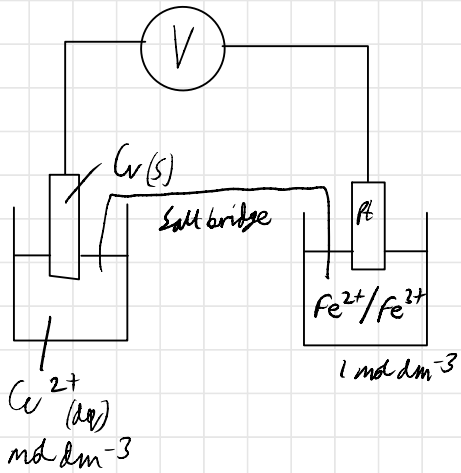
Standard electrode potential



Inert platinum electrode

Standard conditions

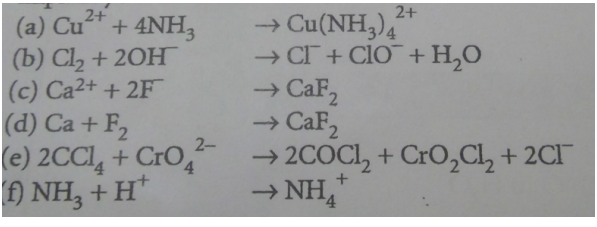
- 298 K
- 100 kPa
- 1.0 mol dm^{-3}



Redox Equations

1. Work out the oxidation number of…
(a) Na in NaOH
(b) Cr in CrO_4^{2-}
(c) O in H_2O_2

2. Are these redox reactions or not?



3. Copper metal can displace silver ions in solution to form silver metal and copper ions. The half equations are: $Cu_{(s)}$ $Cu^{2+}_{(s)} + 2e^-$ $Ag^+_{(ag)} + e^-$ $Ag_{(s)}$
Combine these half equations to write a full, balanced redox equation for the reaction.

4. Dichromate ions are reduced by zinc. The half equations for this reaction are:
$Zn_{(s)}$ $Zn^{2+}_{(aq)} + 2e^-$ $Cr_2O_7^{2-}{}_{(aq)} + H^+_{(aq)} + e^-$
$Cr^{3+}_{(aq)} + H_2O_{(aq)}$
Balance each half equation. Combine these half equations to write a full, balanced redox equation for the reaction.

5. The following reaction occurs between iron and chlorine:

$Cl_{2(g)} + 2Fe^{2+}_{(aq)} \rightarrow 2Cl^-_{(aq)} + 2Fe^{3+}_{(aq)}$

One of the half equations for the reaction is: $Fe^{2+}_{(aq)} \rightarrow Fe^{3+}_{(aq)} + e^-$

Work out the other half equation.

6. Work out both half equations for each of the following reactions:

(a) $CuO + H_2 \rightarrow Cu + H_2O$

(b) $2Na + H_2 \rightarrow 2NaH$

(c) $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$

Eg. Can Cl_2 oxidise Fe^{2+}/Fe^{3+}

$Fe^{3+} + e^- \rightleftharpoons Fe^{2+} + 0.77V$ most -ve

$Cl_2 + 2e^- \rightleftharpoons 2Cl^- + 1.36V$

Cl_2 half equation has more positive value than Fe^{2+}/Fe^{3+} so Cl_2 is reduced.

2. $Cl_2 + 2e^- \rightleftharpoons 2Cl^- + 1.36V$

$MnO_4^- + 8H^+ + 5e^- \rightleftharpoons Mn^{2+} + 4H_2O + 1.52V$

$2Cl^- \rightarrow 5Cl_2 + 2e^-$

$2MnO_4^- + 8H^+ + 5e^- \rightarrow 2Mn^{2+} + 4H_2O$

Feasibility questions

1. $Fe^{3+} + e^- \rightleftharpoons Fe^{2+}$ ←
   0.77 V

   $Br_2 + 2e^- \rightleftharpoons 2Br^-$
   1.09 V
   →

   $2Fe^{2+} + Br_2 \rightarrow 2Br^- + 2Fe^{3+}$ ✓

2. $2I^- \rightleftharpoons I_2 + 2e^-$
   −0.54

   $Br_2 + 2e^- \rightleftharpoons 2Br^-$
   +1.07

   $Br_2 + 2I^- \rightarrow 2Br^- + I_2$ ✓
   solution go from ~~clear~~ orange to brown. ✓

3. $2Cl^- \rightleftharpoons Cl_2 + 2e^-$
   1.36

   $Br_2 + 2e^- \rightleftharpoons 2Br^-$
   1.07

   $Cl_2 + 2Br \rightarrow Br_2 + 2Cl^-$ ✓

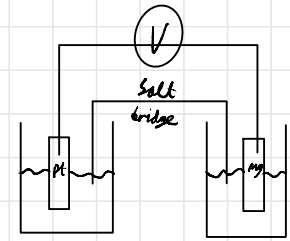
4. $Fe^{3+} + e^- \rightleftharpoons Fe^{2+}$ →
   0.77V

   $Sn^{4+} + 2e^- \rightleftharpoons Sn^{2+}$
   0.15V
   ←

   $2Fe^{3+} + Sn^{2+} \rightarrow 2Fe^{2+} + Sn^{4+}$ ✓
   orange brown to pale green

5. $Zn^{2+} + 2e^- \rightleftharpoons Zn$
   −0.76V
   $Fe^{3+} + e^- \rightleftharpoons Fe^{2+}$
   0.77V

Practice questions



$Fe^{3+} + e^- \rightleftharpoons Fe^{2+}$ $Mg^{2+} + 2e^- \rightleftharpoons Mg$

$1\ mol/dm^3$

298K

100 kPa

ii) electrons in wire, ions in salt bridge

iii) 1.93, Mg more negative

b) i) $Fe^{3+} + e^- \rightleftharpoons Fe^{2+}$

ii) Mg

3. $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$

$\times 5 \quad Fe^{2+} \rightarrow Fe^{3+} + e^-$

$MnO_4^- + 8H^+ + 5Fe^{2+} \rightarrow 5Fe^{3+} + 4H_2O + Mn^{2+}$

	MnO_4^-	$5Fe^{2+}$
Conc	0.02	
mol	0.0005	0.0025
Vol	0.025	

ratio 1 : 5

$0.0025 = \frac{mass}{55.8}$

$mass = 0.1395\,g$ 1 mol (II)

$\frac{0.1395}{3} \times 100 = 4.65\%$

4. $(NH_4)_2SO_4FeSO_4 \cdot nH_2O + kMnO_4 \rightarrow$

8.492g

	Salt	$kMnO_4$
Conc		0.015
Mol		0.0003375
Vol	0.025	0.0225
ratio		

$0.0225 \times 0.015 = 0.0003375$

Unstructured A2 questions

1. 0.11g Ba 100 cm^3

$Ba + 2H_2O_{(l)} \rightarrow Ba(OH) + H_2$

	Ba	$Ba(OH)_2$
Conc		
mol	8.01×10^{-4}	8.01×10^{-4}
vol	$0.1 dm^3$	

$mol = \frac{mass}{M_r} = \frac{0.11}{137.3} = 8.01 \times 10^{-4}$

$n = cV$

$c = \frac{V}{n}$

$8.01 \times 10^{-4} \times (137.3 + 2(16+1)) = 0.1372113$

2. 0.321g

C = 12

1 mol C = 12g

$0.321g \times 6.02 \times 10^{23} = 1.93242 \times 10^{23}$ atoms

3. H_2SO_4

Conc

mol

vol 0.025

$Na_2SO_4 \cdot XH_2O = 322.1 g mol^{-1}$

$23 \times 2 + 32.1 + 16 \times 4 + 18X = 322.1$

$18X = 322.1 - 142.1 = 180$

$X = 10$

4.

Y1 titration 2

1. a) volumetric flask
   b) glass pipette

2. a) unknown solution in burette
   1 ✓ 2, 3 ✗
   D = only 1

3. A

4.

	2NaOH	H_2SO_4
conc	0.144	0.1
mol	0.0036	0.0018
vol	0.025	0.018

$n = cv$ $\quad 0.1 \times 0.018 = 0.0018$

$c = \frac{n}{v}$ $\quad \frac{0.0036}{0.025} = 0.144$

i) 0.0018 mol

ii) 0.144 $mol\,dm^{-3}$

6)

	H_2SO_4	$NaHCO_3$	
conc			
mol			
vol			

5.

	A	NaOH
conc	0.1212	0.120
mol	0.00303	0.00303
vol	0.025	0.02525
	1	1

$0.12 \times 0.02525 = 0.00303$

$\frac{0.00303}{0.025} = 0.1212$

$25 \times 10 = 250$

$0.00303 \times 10 = 0.0303$

$Mr = \frac{mass}{n} = \frac{2.24}{0.0303} = 73.93\ g\,mol^{-1}$

6.7. 24.8g $Na_2S_2O_3 \cdot 5H_2O$

a) i) volumetric flask
ii) glass pipette ✓
ii) burette ✓

b) moles = $\frac{\text{mass}}{Mr}$

$Mr(Na_2S_2O_3 \cdot 5H_2O) = 23\times2 + 32.1\times2 + 16\times3 + 5(18) = 248.2$ ✓

c) $Na_2S_2O_3 \cdot 5H_2O = 24.8g$

$\frac{24.8}{248.2} \approx 0.1$ ✓

d) $0.1 \times 0.0236 = 0.00236$ ✓

e) $S_2O_3^{2-} + I_2$

Analysis of Copper

$$2Cu^{2+} + 4I^- \rightarrow 2CuI + I_2$$

Titration (burette) $2S_2O_3^- + I_2 \rightarrow 2I^- + S_4O_6^{2-}$

$2S_2O_3 \quad 2Cu^{2+}$

$1 \quad : \quad 1$